

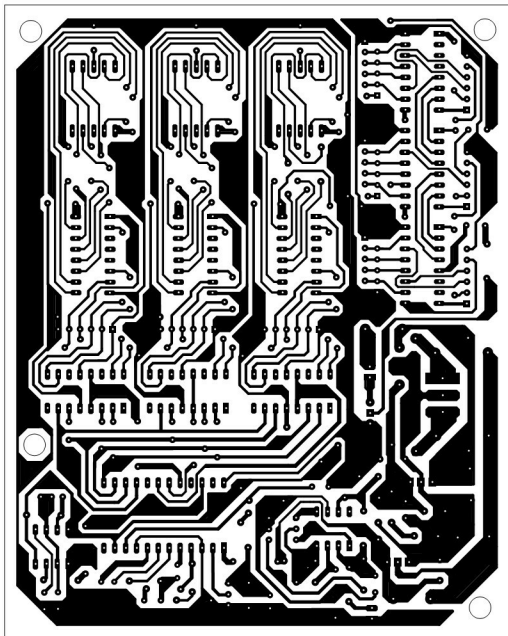


Fig. 5: An actual-size, single-side PCB for the digital temperature controller using thermocouple

perature controller is shown in Fig. 5 and its component layout in Fig. 6. Suitable connectors are provided on PCB for comparators, thumbwheel switches and relay. Assembling the circuit on a PCB minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Use bases to avoid damage to ICs due to direct soldering and over-heating.

Testing and calibration

For testing and calibration of the digital temperature controller, you need a good-quality, 4-digit millimeter capable of reading microvolts to 100V DC

in different ranges and a stable and accurate voltage injector capable of injecting microvolts to volts in variable mode as per the requirement of the K-type thermocouple.

After assembling and wiring the circuit, connect 230V, 50Hz mains supply to the primary winding of the transformer of the power supply unit and check +5V and -5V supplies. If both the supplies are okay, switch off the supply by opening switch S1 and insert appropriate ICs into IC bases on the PCB.

Zero setting. Short the inverting and non-inverting pins (pins 2 and 3) of op-amp μ A741 and connect the digi-

PARTS LIST

Semiconductors:

IC1	- μ A741 operational amplifier
IC2	- ICL7135 4½-digit ADC
IC3	- NE555 timer
IC4-IC6	- 74LS75 quad latch
IC7-IC9	- 74LS48 BCD to 7-segment decoder
IC10-IC12	- 74LS85 magnitude comparator
IC13	- 7805, +5V regulator
IC14	- 7905, -5V regulator
T1	- BC557 pnp transistor
ZD1	- 3.3V zener diode
D1	- 1N4148 switching diode
D2-D9	- 1N4007 rectifier diode
LED1	- 5mm LED
DIS1-DIS3	- LTS543 common-cathode 7-segment display

Resistors (all ¼-watt, $\pm$ 5% carbon):

R1, R2	- 10-ohm
R3, R5	- 100-kilo-ohm
R4	- 10-kilo-ohm
R6	- 27-ohm
R7, R9	- 1-mega-ohm
R8, R11, R12	- 1-kilo-ohm
R10, R13	- 150-ohm
R14-R34	- 330-ohm
R35	- 2.2-kilo-ohm
R36	- 470-ohm
RNW1-RNW3	- 10-kilo-ohm resistor network
VR1	- 50-kilo-ohm preset
VR2, VR4	- 10-kilo-ohm preset
VR3	- 2-kilo-ohm preset

Capacitors:

C1, C2	- 470 μ F, 16V electrolytic
C3	- 0.47 μ F ceramic disk
C4, C8	- 1 μ F, 25V electrolytic
C5, C6, C10	- 0.01 μ F ceramic disk
C7	- 10 μ F, 16V electrolytic
C9	- 470pF ceramic disk
C11, C12	- 1000 μ F, 25V electrolytic
C13, C14	- 470 μ F, 16V electrolytic
C15, C16	- 0.1 μ F ceramic disk

Miscellaneous:

X1	- 230V AC primary to 9V-0-9V, 1A secondary transformer
S1	- On/off switch
TWS1-TWS3	- BCD thumbwheel switch
RL1	- 12V, 1C/O relay
F1	- 1A fuse
TC	- K-type thermocouple
CON1-M	- 6-pin berg strip male connector
CON3-M	- 6-pin berg strip male connector
CON1-F	- 6-pin berg strip female connector
CON3-F	- 6-pin berg strip female connector

tal millivoltmeter to pin 6 of μ A741. Set the reading to '000' on multimeter by varying preset VR2 (connected to pins 1 and 5). When the display shows 000,